

University of Florida

Ph.D in Computer Science

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Machine Learning, A.I. Fundamentals, Natural Language Processing, Maths for Intelligent Systems

Gainesville, FL
May 2028

Technical Skills

Languages: C++, Python, Java, JavaScript, MATLAB, SQL

Frameworks & Tools: GitHub, Node.js, Android Studio, Jupyter Notebook

Research & Work Experience

City Guarding of Buildings With Arbitrary Bases And Limited View

January 2025 - Present

- Extended a previous conjecture from computational geometry on the minimum number of cameras required to guard cities composed of rectangular prisms using 180°FOV cameras to buildings of arbitrary polygon bases
- Reformulated prior lemmas geometric constraints and analyzing sufficiency and necessity conditions

Teaching Assistant - University of Florida

August 2025 - Present

- Assisted over 300 students by hosting weekly office hours to provide academic support and assist with topics of algorithm design, complexity analysis, and programming language concepts
- Provided constructive feedback on code quality, quizzes, and exams to improve their understanding and performance.
- Setup additional study sessions prior to quizzes/exams to better assist students in the course

Impact of A.I. in Large Classroom Setting - Accepted With Minor Revisions

January 2024 - November 2024

- Conducted a study to measure if A.I. technologies can increase student engagement in large classroom settings
- Performed data and conversation analysis of over 200 students' conversations to observe how well students were engaged in learning. Observed students reported higher satisfaction and engagement compared to control group
- Submitted a paper which has detailing the background on AI in education, how the study was conducted, and the significance of the data

Technical Projects

Scratch Implementation of a Feedforward Neural Network for Binary Classification (Python, Numpy, Matplotlib)

March 2026 - April 2026

- Achieved 96% classification accuracy on a dataset of 1000 points by building and training a custom neural network (2–3–1 architecture) from scratch using sigmoid activation and backpropagation over 10,000 epochs
- Developed a binary classifier to detect whether 2D points lie within a triangular region, improving predictive performance by manually implementing forward propagation, loss computation, and gradient-based weight updates
- Evaluated model generalization by expanding the input space beyond the training distribution, identifying a drop to 74% accuracy and analyzing limitations in learned decision boundaries

Signify ASL Interpreter and Study Model (Android Studio, Kotlin, SQL)

January 2025 - April 2025

- Collaborated in a team to design and develop a mobile application for learning American Sign Language (ASL)
- Designed an implemented an SQLite relational database schema storing and querying user information, achievements, and discussions
- Enabled seamless app updates by managing database version control through onUpgrade(), ensuring new columns could be added without disrupting user data
- Increased development efficiency by creating a debug page within the app that allowed the team to test database functionality before implementation, reducing integration errors

Modeling Lead Bioaccumulation in Human Body from Chocolate Consumption

October 2024 - November 2024

- Developed a two-compartment differential equation model describing the circulation of lead between blood and bone tissue in the human body
- Analyzed long-term dynamics using phase plane and eigenvalue analysis, showing the system converges to a stable equilibrium representing steady state lead levels

Leadership and Professional Development

UF National Society of Black Engineers: Executive Board

May 2024 – May 2025

Secretary

- Coordinated monthly chapter meetings and managed internal communications to ensure smooth information flow across the organization
- Delegated tasks to members for event promotion and engagement (flyer creation, photography, website updates, and social media posts)
- Scheduled and reserved rooms for chapter events, optimizing event logistics and attendance

Awards

BRIDGE Scholarship Recipient, Cum Laude Graduate, Meritorious Award SCUDEM 2024, Successful Participant MCM/ICM 2024, Dean's List Spring 2023, Successful Participant MCM/ICM 2023, Meritorious Award SCUDEM 2022